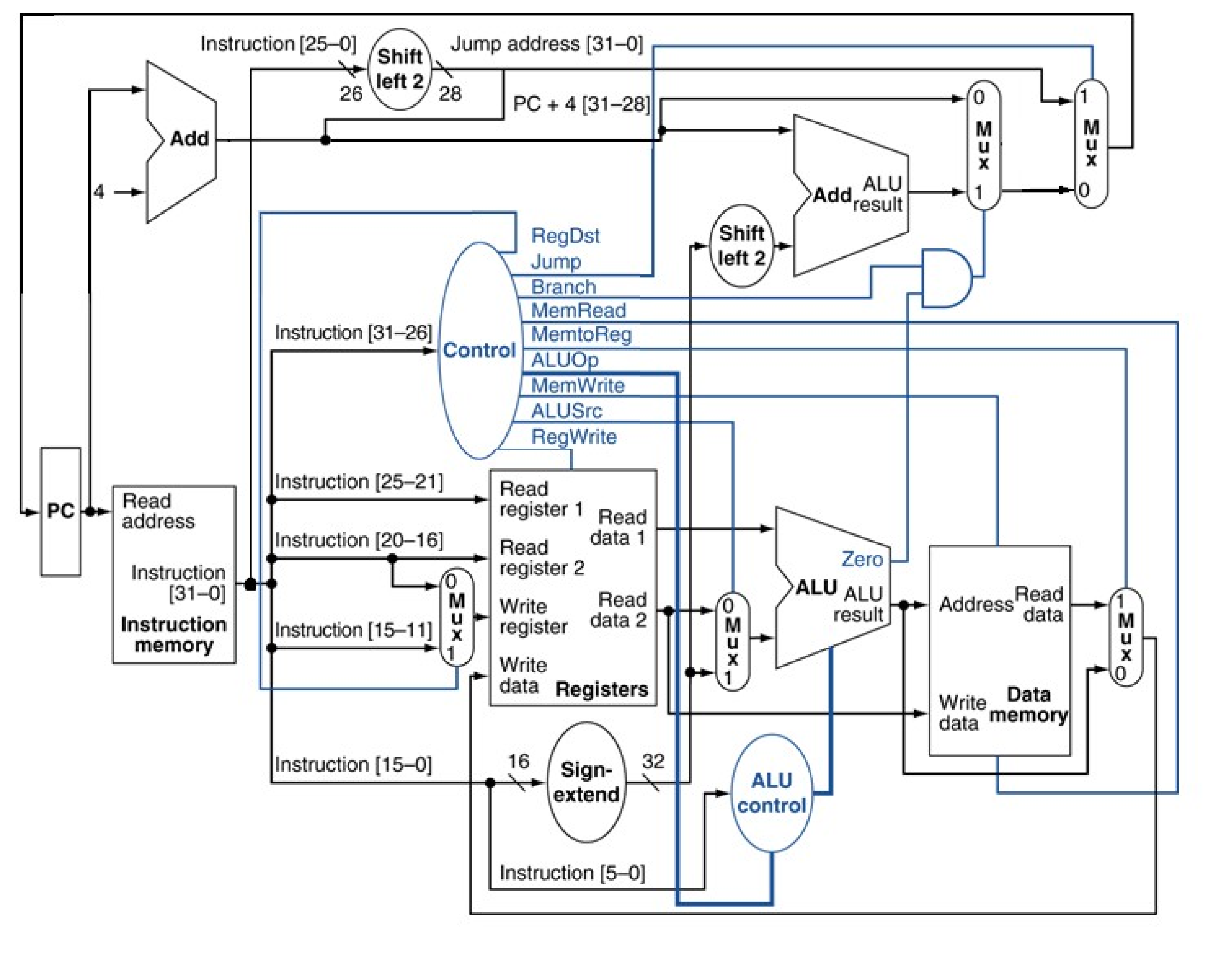
CDA 3103

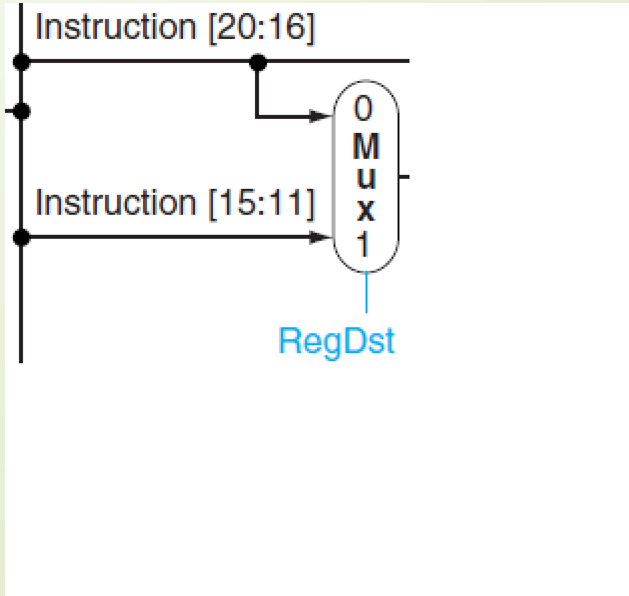
Recitation 10 – Control Signals

MIPS Single Cycle Datapath



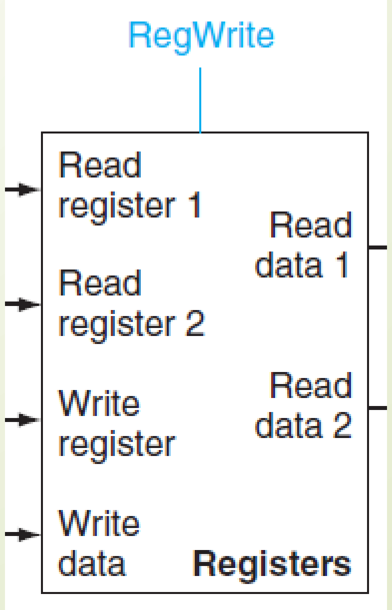
Control Signals – Register Destination

- RegDst
  - Deasserted: the register destination number for the write register comes the rt field (bits 20:16)
  - Asserted: the register destination number for the write register comes the rd field (bits 15:11)



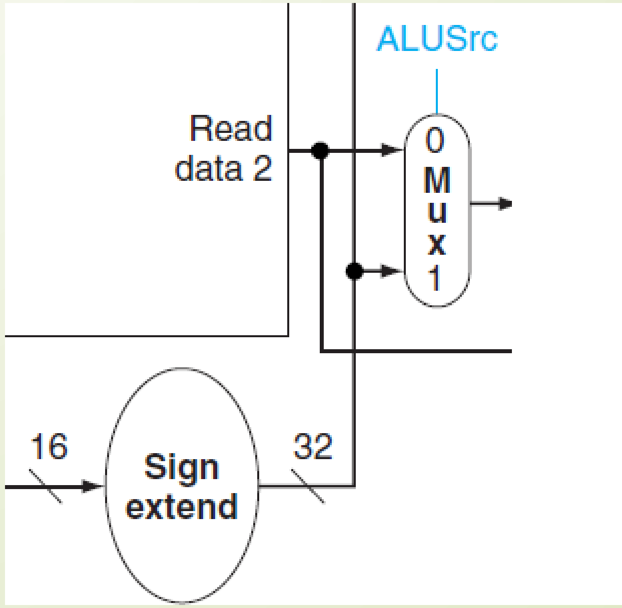
Control Signals – Register Write

- RegWrite
  - Asserted: the register destination on the write register input is written with the value on the write data input



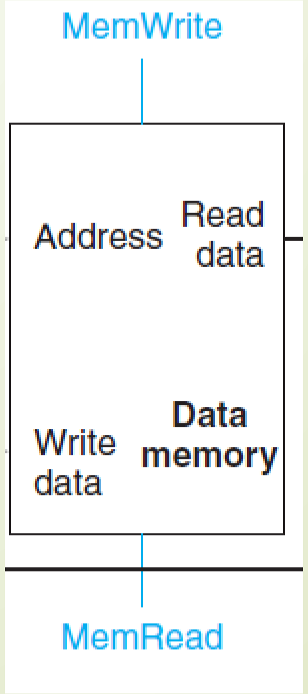
Control Signals – ALU Source

- ALUSrc
  - Deasserted: The second ALU operand comes from the second register file output
  - Asserted: The second ALU operand is the sign-extended lower 16 bits of the instruction



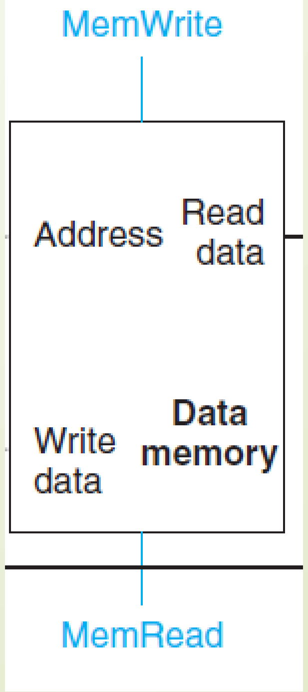
Control Signals – Memory Read

- MemRead
  - Asserted: Data memory contents designated by the address input are put on the read data output.



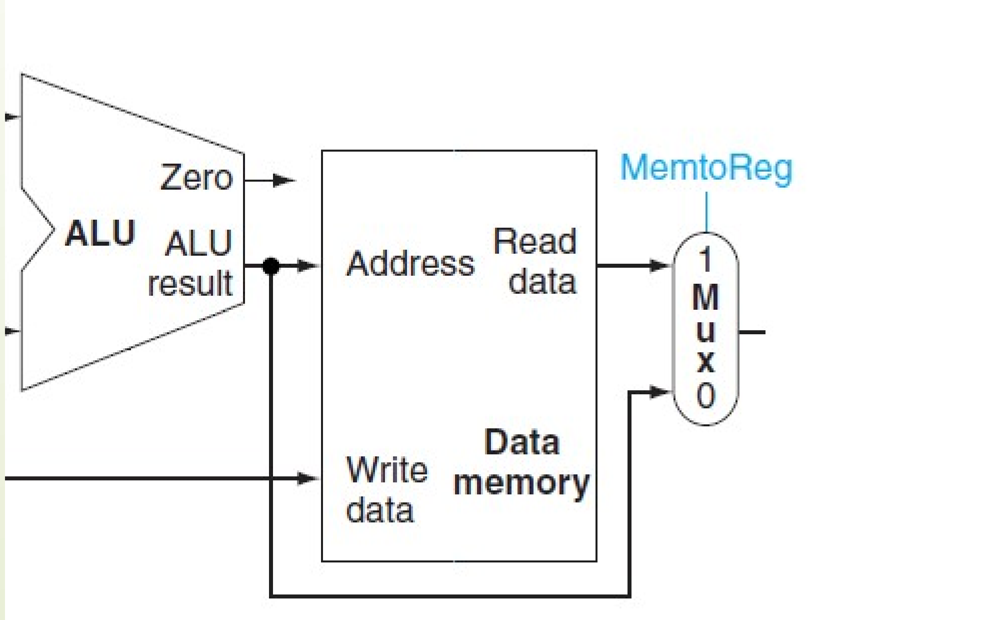
Control Signals – Memory Write

- MemWrite
  - Asserted: Data memory contents designated by the address input are replaced by the value on the write data input.



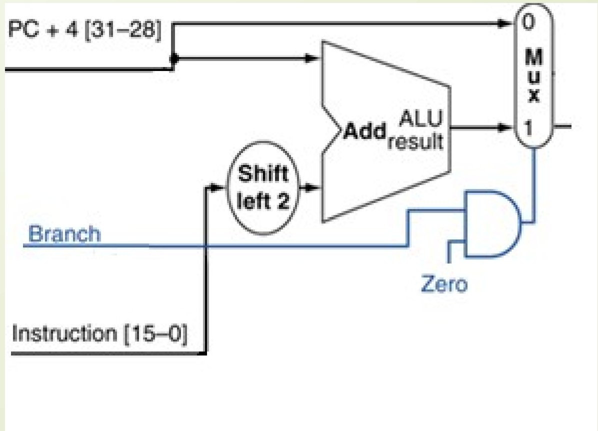
Control Signals – Memory to Register

- MemtoReg
  - Deasserted: The value fed to the register write data input comes from the ALU
  - Asserted: The value fed to the register write data input comes from the data memory.



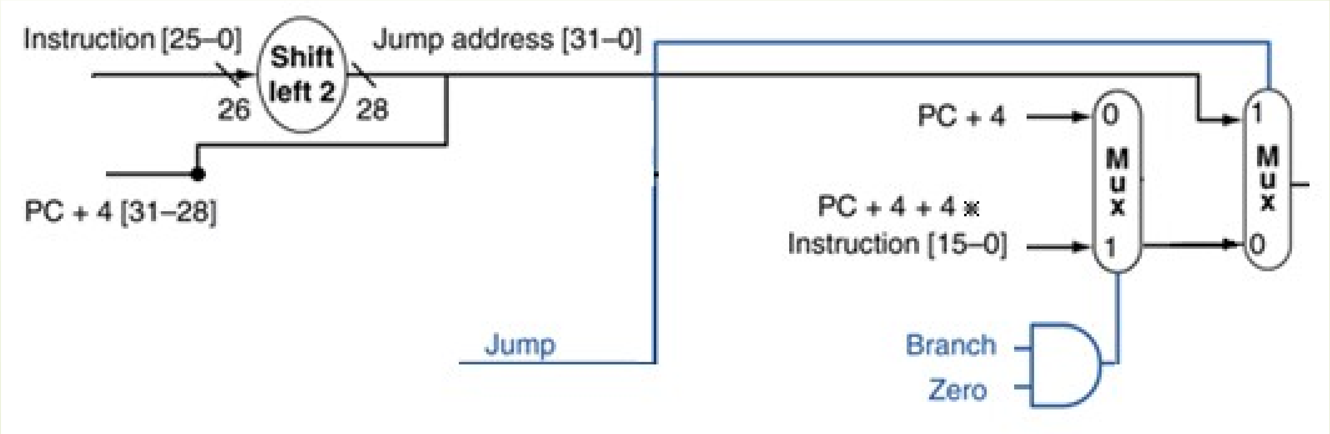
Control Signals – Branch

- Branch on Equal
  - Deasserted: PC = PC + 4
  - Asserted: Check Zero output of the ALU
    - When Zero is asserted use the branch target address. PC = PC + 4 + 4*Instruction[15-0]



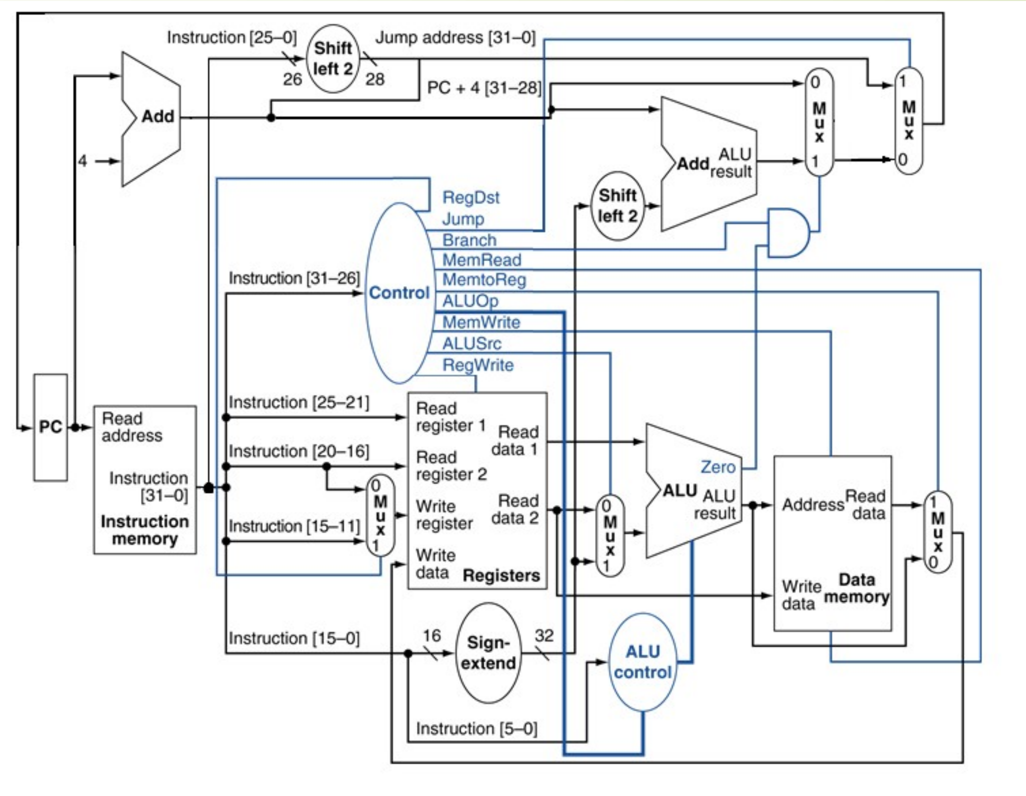
Control Signals – Jump

- Jump
  - Deasserted: PC is either PC + 4 or the branch target
  - Asserted: PC = PC + 4[31-28] : Instruction[25-0] * 4

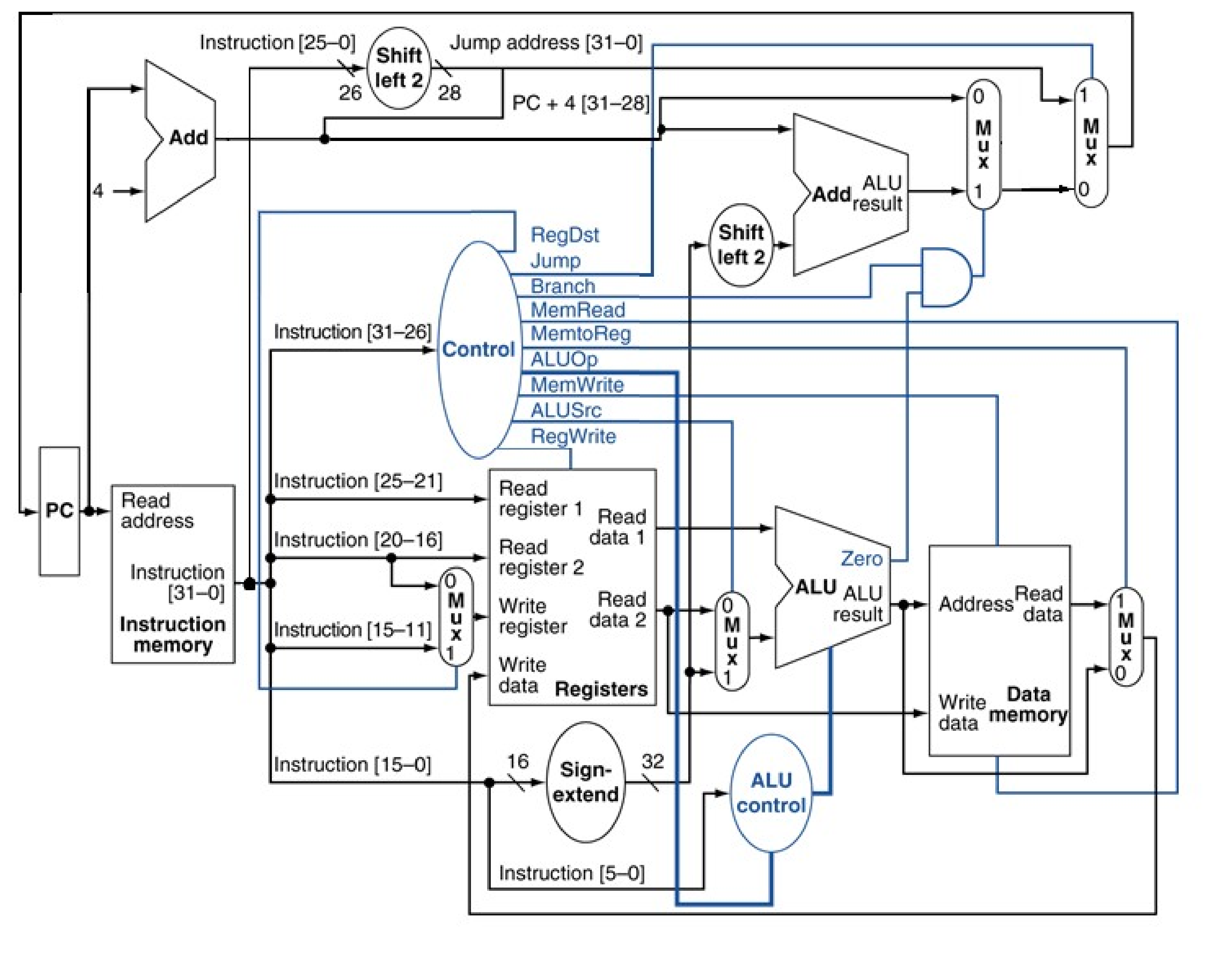


Setting Control Signals

- Consider the single cycle datapath. What would be the values of the control signals for the following instructions: add, addi, sw, beq, j?

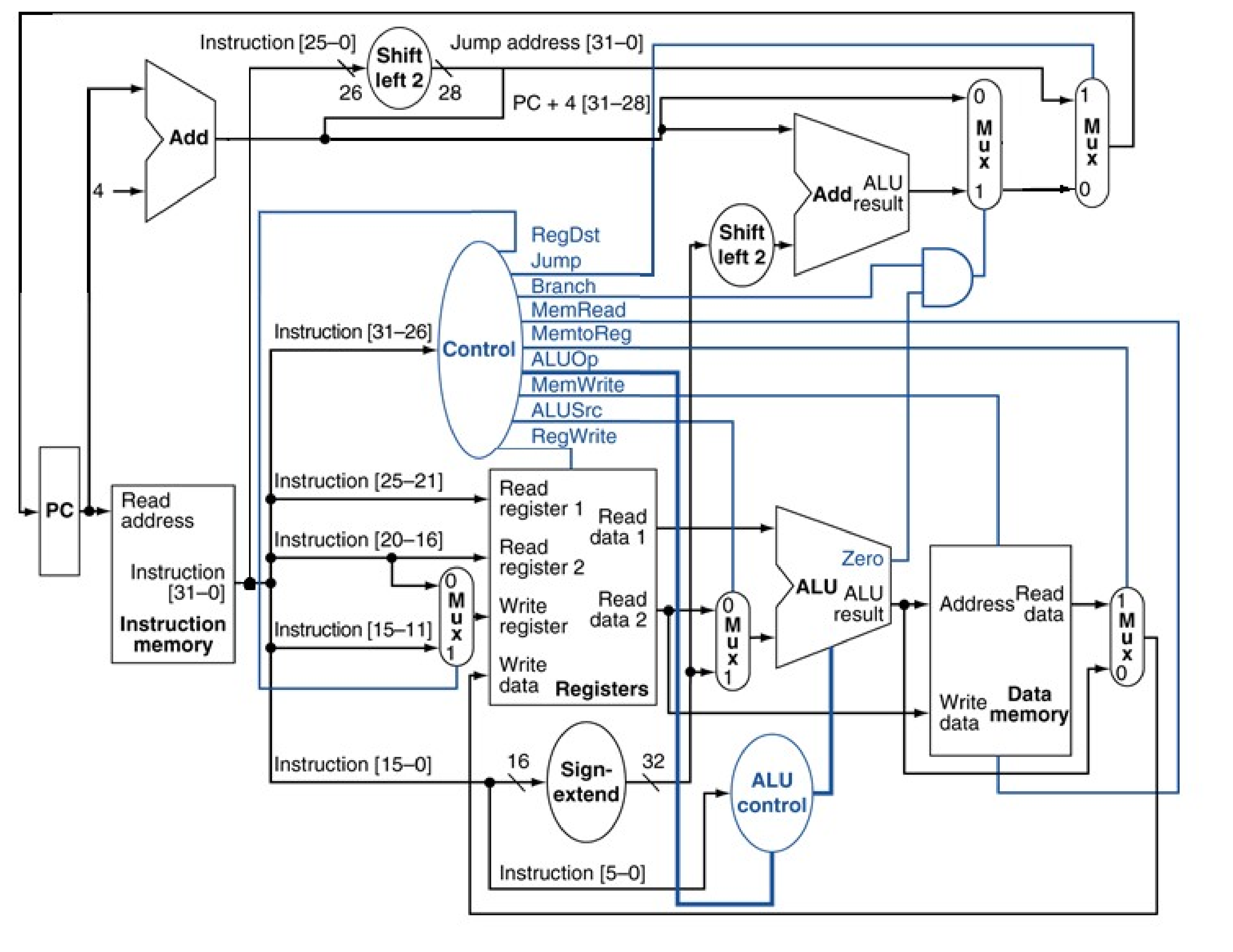


Setting Control Signals: Add



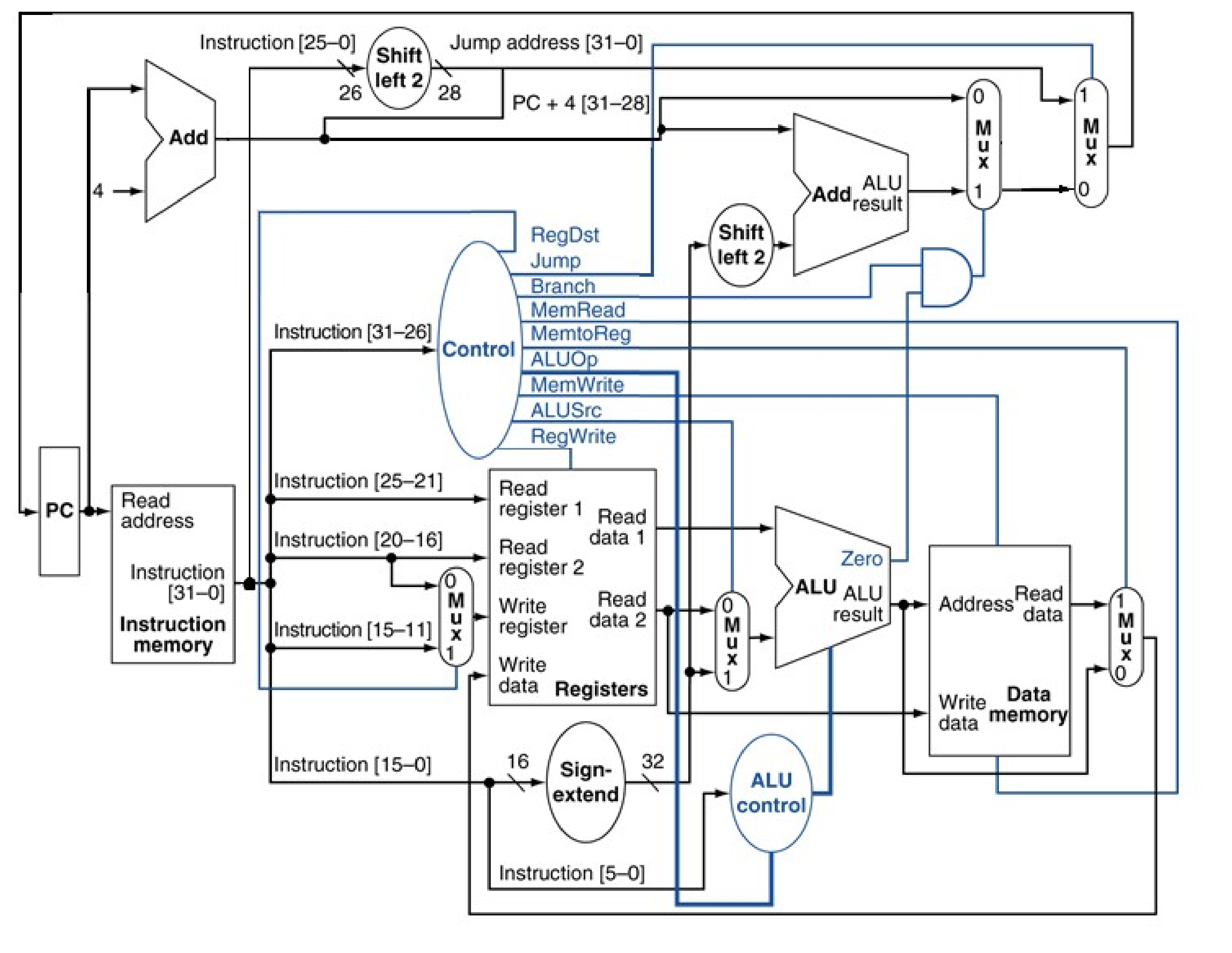
- RegDst: 1
- Jump: 0
- Branch: 0
- MemRead: 0
- MemtoReg: 0
- ALUOp: “add”
- MemWrite: 0
- ALUSrc: 0
- RegWrite: 1

Setting Control Signals: Addi



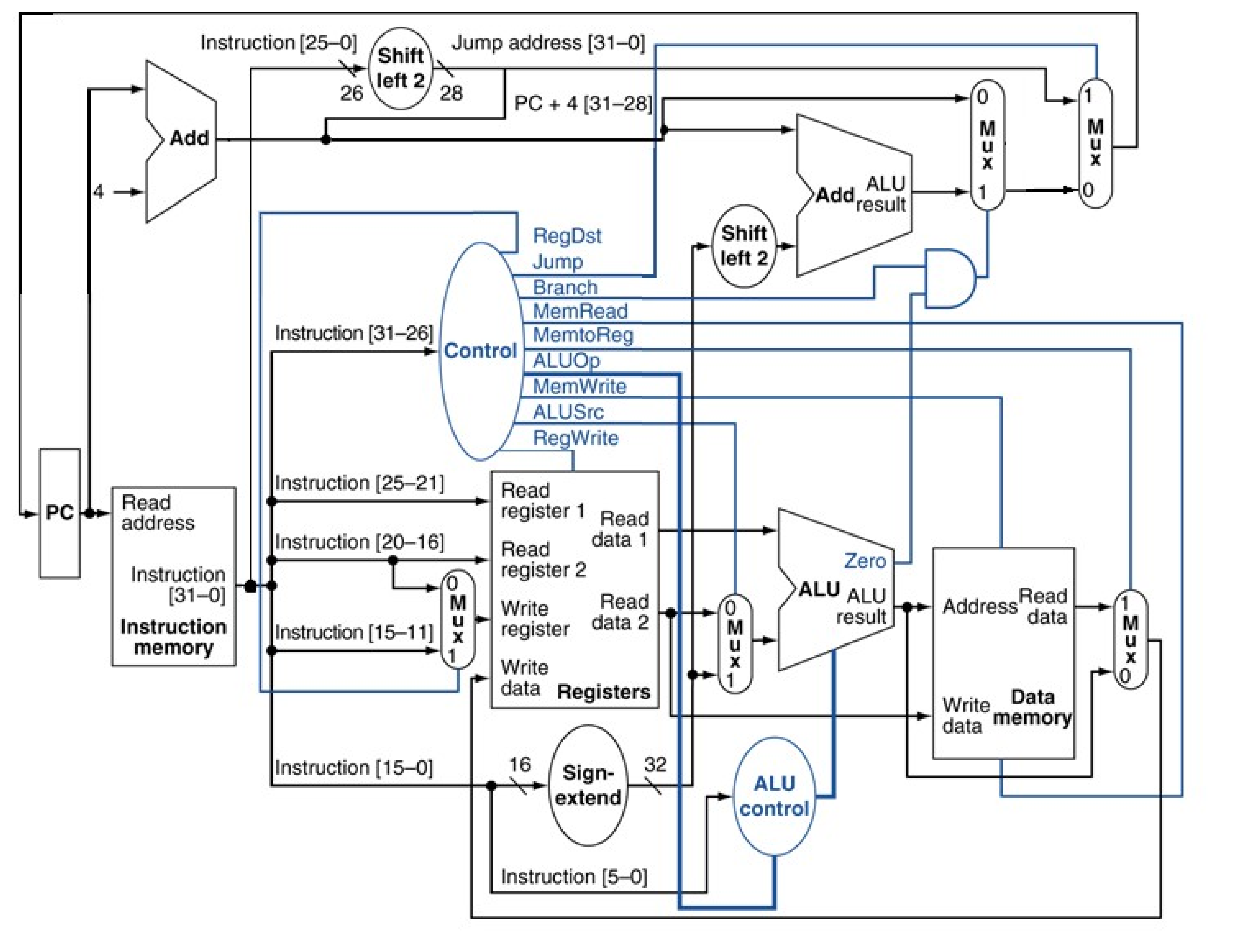
- RegDst:
- Jump:
- Branch:
- MemRead:
- MemtoReg:
- ALUOp:
- MemWrite:
- ALUSrc:
- RegWrite:

Setting Control Signals: Store Word



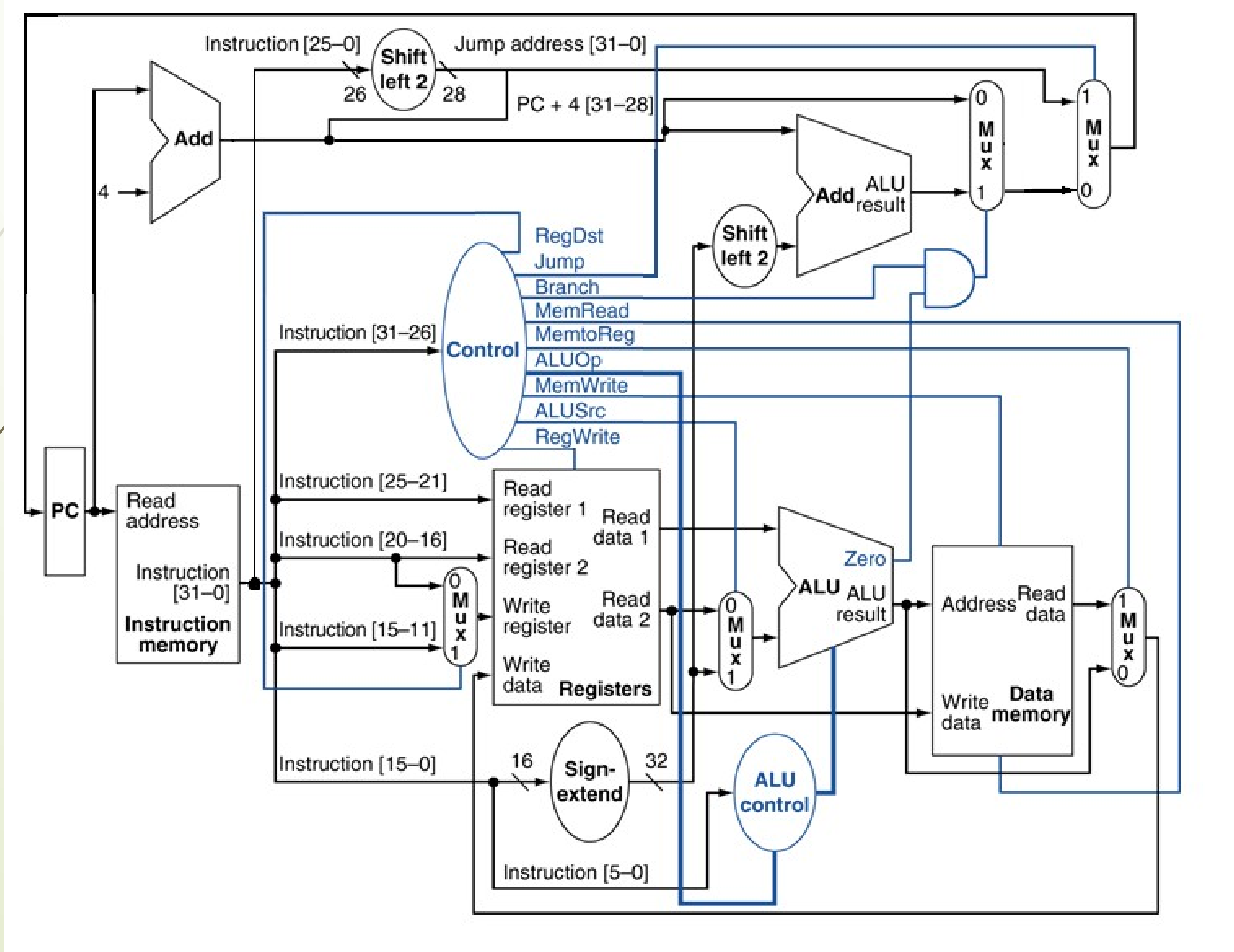
- RegDst:
- Jump:
- Branch:
- MemRead:
- MemtoReg:
- ALUOp:
- MemWrite:
- ALUSrc:
- RegWrite:

Setting Control Signals: BEQ



- RegDst:
- Jump:
- Branch:
- MemRead:
- MemtoReg:
- ALUOp:
- MemWrite:
- ALUSrc:
- RegWrite:

Setting Control Signals: Jump



- RegDst:
- Jump:
- Branch:
- MemRead:
- MemtoReg:
- ALUOp:
- MemWrite:
- ALUSrc:
- RegWrite: